

STAT 707: Final Project

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A dark blue diagonal gradient bar that starts from the bottom left corner and extends towards the top right corner, covering the lower half of the slide.

Data Preprocessing

PoolQC	1055
MiscFeature	1021
Alley	987
Fence	849
MasVnrType	636
FireplaceQu	499
LotFrontage	188
GarageType	58
GarageFinish	58
GarageQual	58
GarageYrBlt	58
GarageCond	58
BsmtFinType2	31
BsmtExposure	31
BsmtCond	30
BsmtFinType1	30
BsmtQual	30
MasVnrArea	6

- Na counts shown
- Na does not mean “missing value”, rather it typically refers to “missing feature”
- Ex: if MiscFeature == “na” then there are no miscellaneous features

Date Preprocessing

```
['MSZoning' 'MSZoning_C (all)']  
['Street' 'Street_Grvl']  
['LotShape' 'LotShape_IR1']  
['LandContour' 'LandContour_Bnk']  
['Utilities' 'Utilities_AllPub']  
['LotConfig' 'LotConfig_Corner']  
['LandSlope' 'LandSlope_Gtl']  
['Neighborhood' 'Neighborhood_Blmngtn']  
['Condition1' 'Condition1_Artery']  
['Condition2' 'Condition2_Artery']  
['BldgType' 'BldgType_1Fam']  
['HouseStyle' 'HouseStyle_1.5Fin']  
['RoofStyle' 'RoofStyle_Flat']  
['RoofMatl' 'RoofMatl_ClyTile']  
['Exterior1st' 'Exterior1st_AsbShng']  
['Exterior2nd' 'Exterior2nd_AsbShng']  
['ExterQual' 'ExterQual_Ex']  
['ExterCond' 'ExterCond_Ex']  
['Foundation' 'Foundation_BrkTil']  
['Heating' 'Heating_Floor']  
['HeatingQC' 'HeatingQC_Ex']  
['CentralAir' 'CentralAir_N']  
['Electrical' 'Electrical_FuseA']  
['KitchenQual' 'KitchenQual_Ex']  
['Functional' 'Functional_Maj1']  
['PavedDrive' 'PavedDrive_N']  
['SaleType' 'SaleType_COD']  
['SaleCondition' 'SaleCondition_Abnorml']]
```

Reference level key

```
['Alley',  
 'MasVnrType',  
 'BsmtQual',  
 'BsmtCond',  
 'BsmtExposure',  
 'BsmtFinType1',  
 'BsmtFinType2',  
 'FireplaceQu',  
 'GarageType',  
 'GarageFinish',  
 'GarageQual',  
 'GarageCond',  
 'PoolQC',  
 'Fence',  
 'MiscFeature']
```

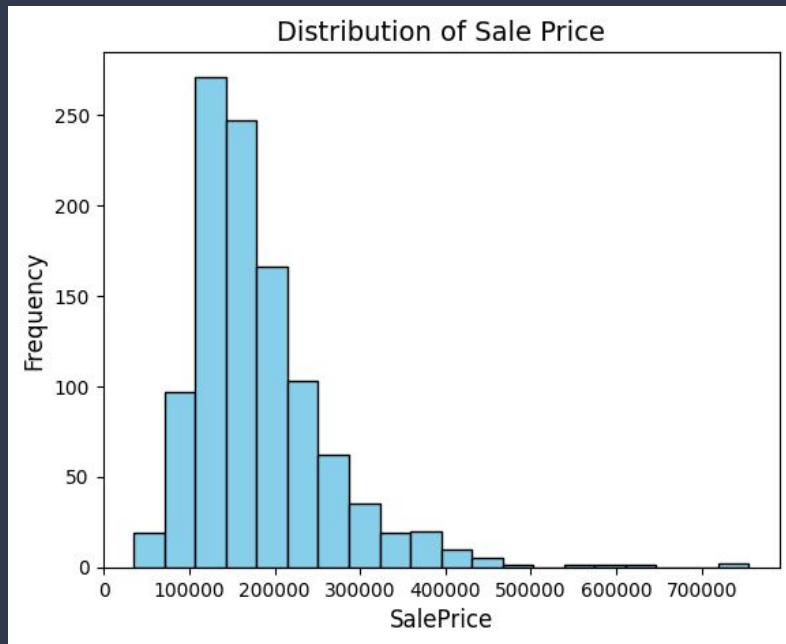
Reference level "na"

- Quantitative variables with missing values were replaced with column median
- Categorical variables were encoded with c-1 dummy variables, with missing values (na) as the reference level if they exist

Data Preprocessing (Ordinal Variables)

- For variables like BsmtQual or ExterCond, which have five levels
 - poor, fair, typical, good, excellent
- A couple ways of encoding this
 - (1) use 4 dummy variables (or 5 if na is present)
 - cost: you lose order
 - (2) make it quantitative: poor = 1, ..., excellent = 5 (with na = 0)
 - cost: assumes “distance” between levels is the same
- Other ordinal variables can be handled similarly

Histogram of Sale Price



- Right Skewed
- Common Associated Problems:
 - Residuals often “inherit” non-normality of response
 - Heteroscedasticity
 - Non-linear relationships

Examining the distribution of our response variable.

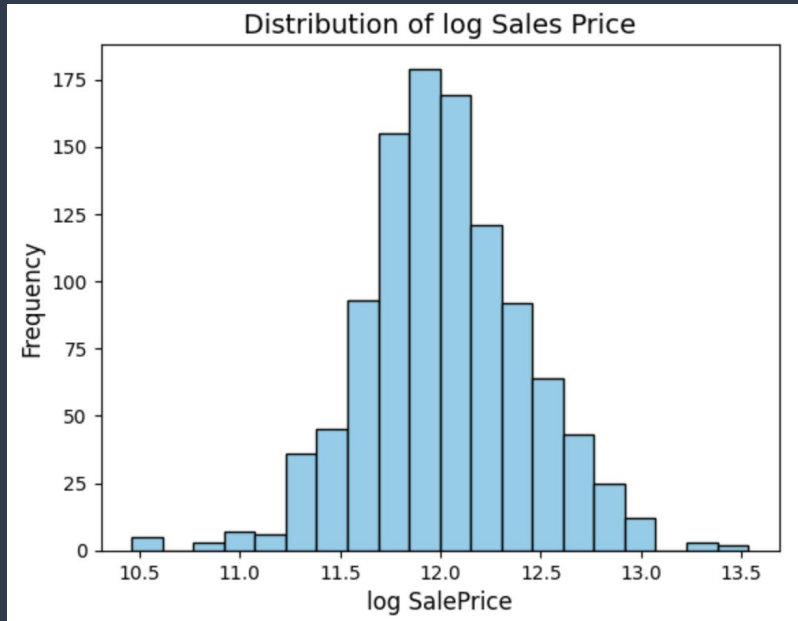
QQ Plot of Sale Price



Very not normal!

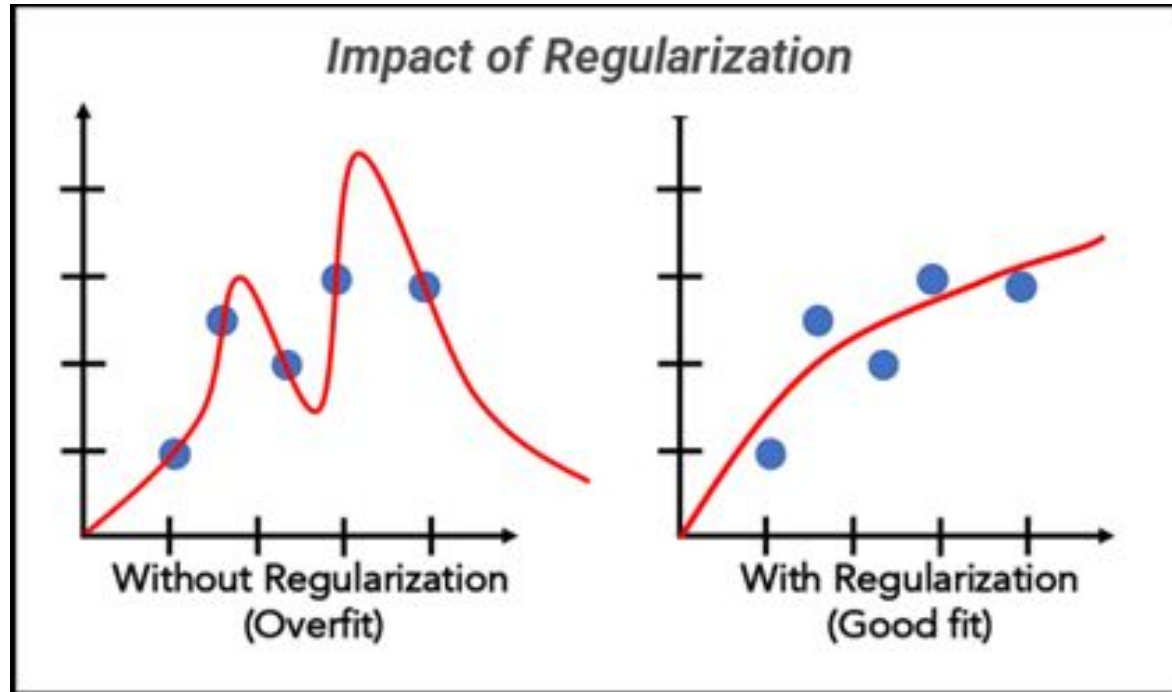
- Constructing QQ plot
- Significance of Linear QQ plot
- Interpretation of QQ plot: Slope, Intercept

Histogram of Log Price



- The $\log(\text{response})$ looks much more normal, which is a good sign for our model assumptions

Picture to motivate topic



How do they work? (LASSO & Ridge)

Different types of regularization (L1 and L2)

$$\sum_{i=1}^M (y_i - \hat{y}_i)^2 = \sum_{i=1}^M \left(y_i - \sum_{j=0}^p w_j \times x_{ij} \right)^2 + \lambda \sum_{j=0}^p w_j^2$$

Cost function for ridge regression

$$\sum_{i=1}^M (y_i - \hat{y}_i)^2 = \sum_{i=1}^M \left(y_i - \sum_{j=0}^p w_j \times x_{ij} \right)^2 + \lambda \sum_{j=0}^p |w_j|$$

Cost function for Lasso regression

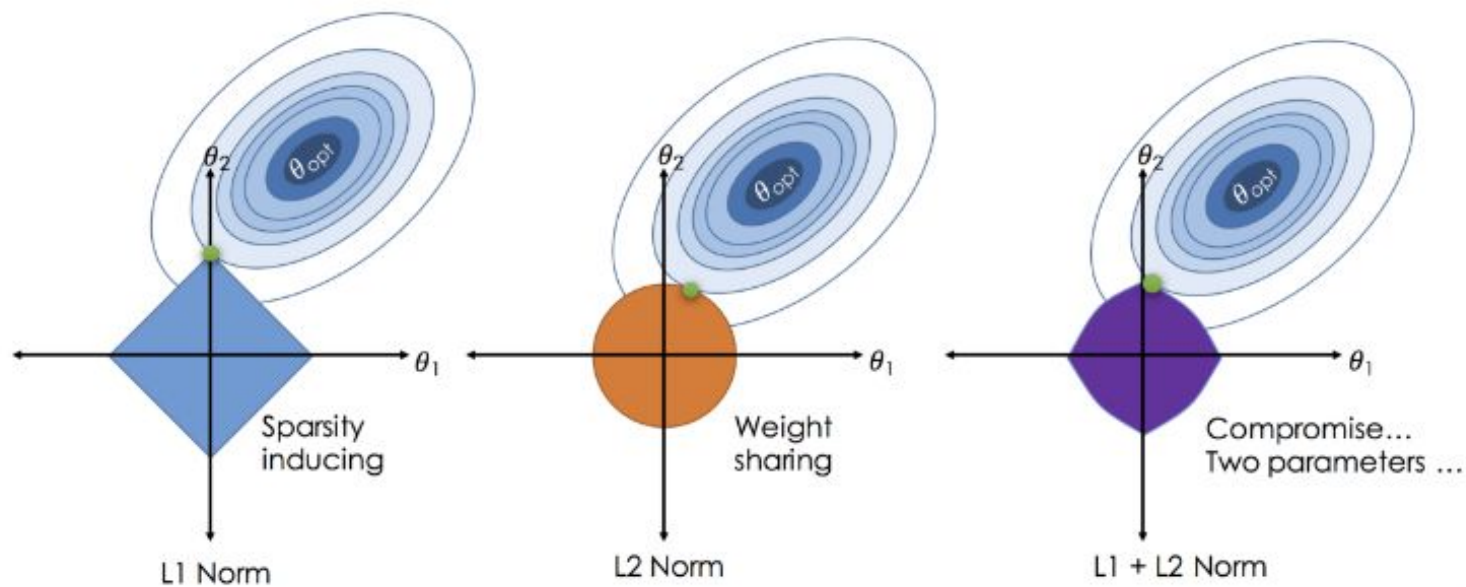
How do they work? (Elastic Net)

$$\text{Loss} = \sum_{i=1}^n (y_i - \hat{y}_i)^2 + \alpha_1 \sum_{i=1}^n |w_i| + \alpha_2 \sum_{i=1}^n w_i^2$$

Where,

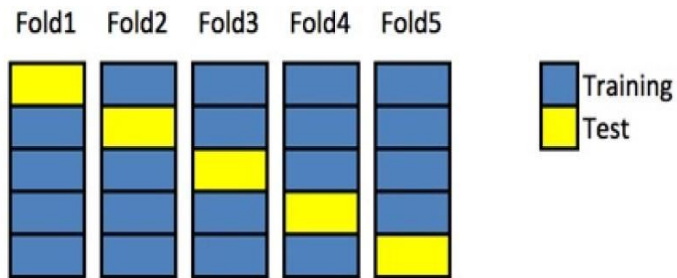
- y_i is actual value
- $\hat{y}_i$ is predicted value
- α_1 penalty of lasso regression
- α_2 penalty of ridge regression
- w_i weights assigned

How do they work? (LASSO, ridge, elastic net)



Model #1:

LASSO & Ridge (LASSO hyperparameter tuning)



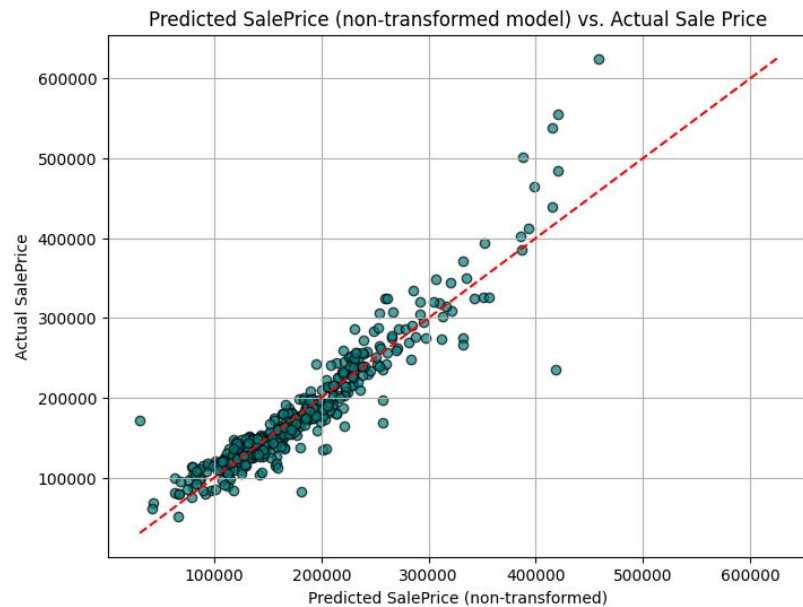
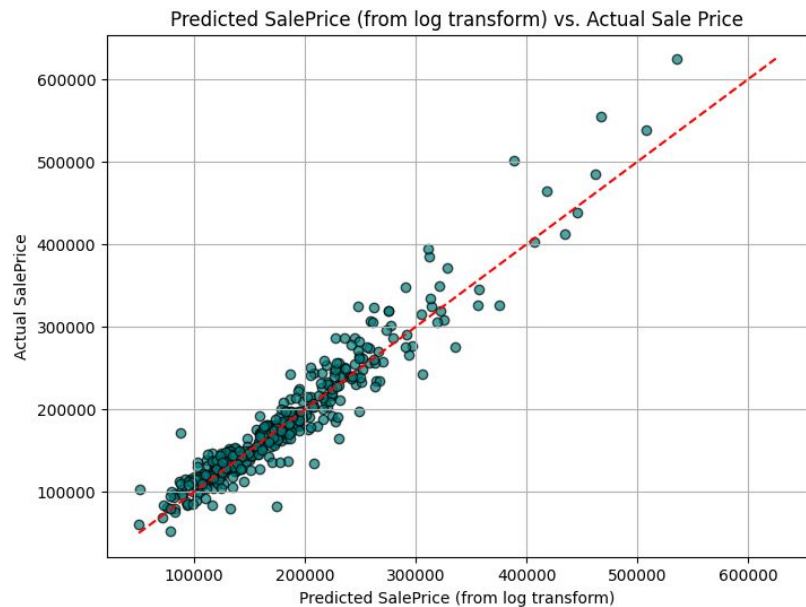
5-fold Cross Validation

1. Separate data into test/training sets with k-groups and k-parts.
2. Build the model in the training sets and score against the test set.
3. Optimize model for lowest MSE.

***Note:** Data was scaled prior to model fitting.

```
param_grid = {'alpha': [0.001, 0.01, 0.1, 1, 10, 100, 1000, 10000, 100000, 1000000, 10000000, 100000000]}
```

Model #1: LASSO and Ridge (LASSO Predictions)



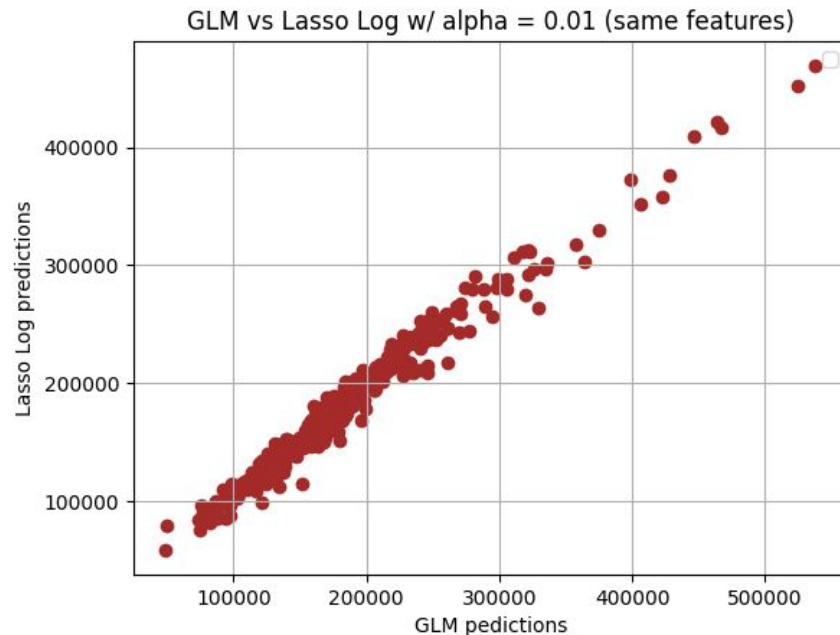
\$400,000

Model #1: LASSO and Ridge (LASSO Evaluation)

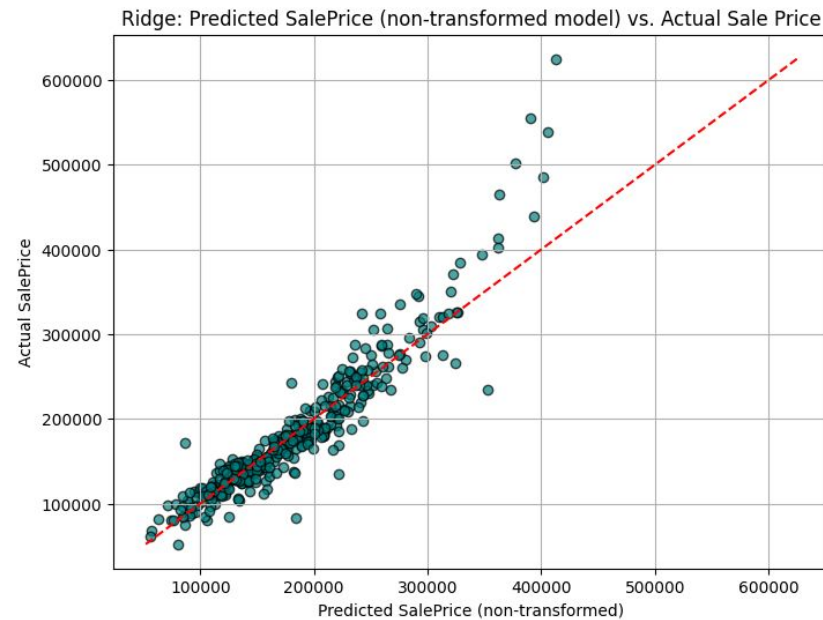
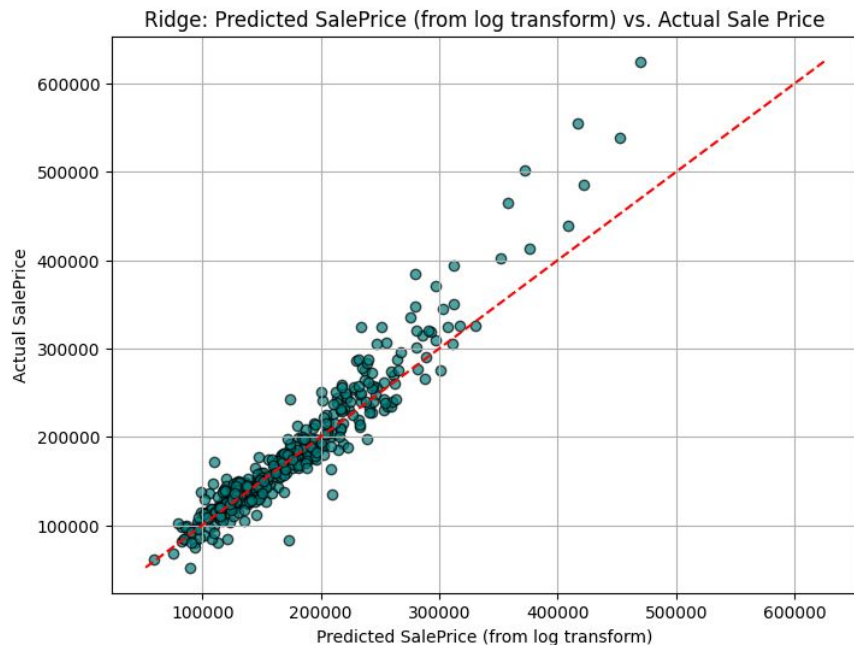
	Log-transformed Model	Non-transformed Model
alpha	0.01	1000
# of unique features	182	87
# of shared features	81	
Bias = $\frac{1}{n} \sum_{i=1}^n (\hat{Y}_i - Y_i)$	-3,773	-1,618
Maximum Deviation = $Max\{ \hat{Y}_i - Y_i \}$	112,950	183,622
Mean Absolute Deviation = $\frac{1}{n} \sum_{i=1}^n \hat{Y}_i - Y_i $	15,521	17,374
Mean Squared Error = $\frac{1}{n} \sum_{i=1}^n (\hat{Y}_i - Y_i)^2$	520,856,703	744,321,833

Model #1: LASSO & Ridge (LASSO/Ridge Variable Selection)

- Alpha = 0.01
- Sparsity vs Shrinkage
- Multicollinearity
- Overfitting



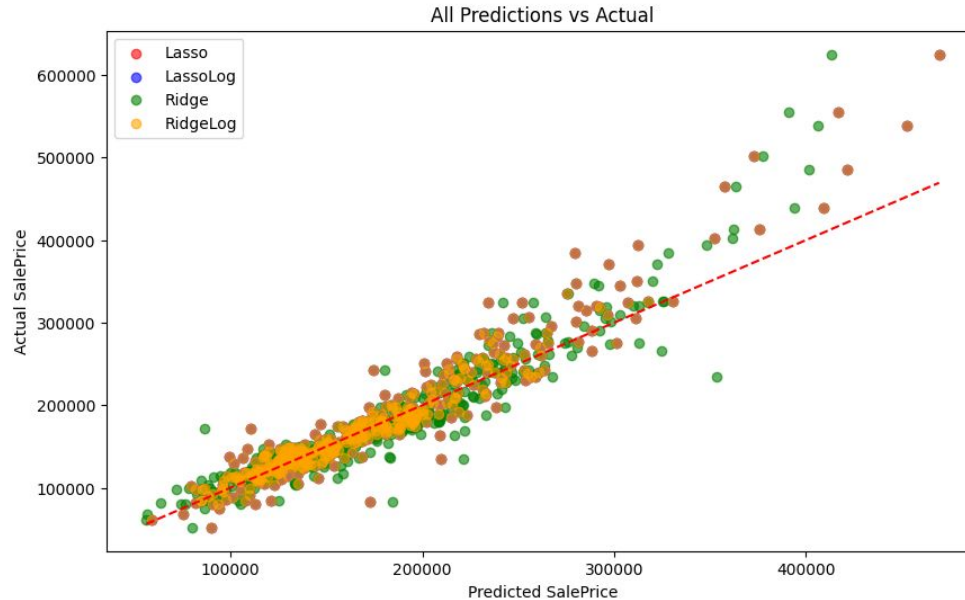
Model #1: LASSO & Ridge (Ridge Predictions)



Model #1: LASSO and Ridge (Ridge Evaluation)

	Log-transformed Model	Non-transformed Model
alpha	1000	1000
5 Largest coefficients	OverallQual, GrLivArea, TotRmsAbvGrd, X1stFlrSF, GarageCars	OverallQual, GrLivArea, PoolQC_Ex , TotRmsAbvGrd, BsmtQual_Ex
Correlation between coefficients	0.985	
Bias = $\frac{1}{n} \sum_{i=1}^n (\hat{Y}_i - Y_i)$	-5,982	-1,618
Maximum Deviation = $Max\{ \hat{Y}_i - Y_i \}$	155,587	211,996
Mean Absolute Deviation = $\frac{1}{n} \sum_{i=1}^n \hat{Y}_i - Y_i $	16,217	16,945
Mean Squared Error = $\frac{1}{n} \sum_{i=1}^n (\hat{Y}_i - Y_i)^2$	658,400,440	767,772,527

Model #1: LASSO & Ridge (4-way comparison)



Under prediction of most expensive houses from all models

Model #1: LASSO & Ridge (Conclusion)

Other related models:

- Elastic Net ($1 < p < 2$)
- Bridge ($p < 1$)
- Group LASSO
- Using Interaction and Polynomial terms with LASSO

Limitations:

- Possible heteroskedasticity
- Poor predictions over \$400k
- LASSO convergence was slow for SalePrice but fast for $\log(\text{SalePrice})$
- Penalized regression purposely adds bias to the model

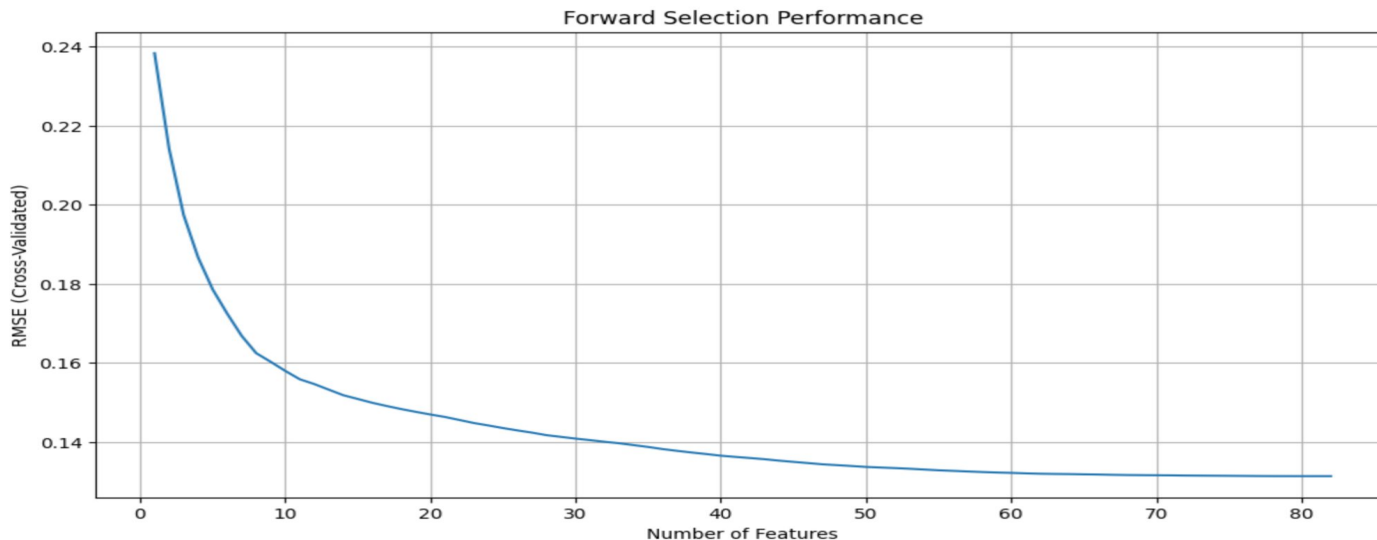
Model #1: LASSO & Ridge (4-way comparison)

	LASSOLog	LASSO	Ridge	RidgeLog
lambda	0.01	1000	1000	1000
Bias	-3,773	-1,618	-1,619	-5,982
Maximum Deviation	112,950	183,622	211,996	155,587
Mean Absolute Deviation	15,521	17,374	16,945	16,217
Mean Squared Error	520,856,703	744,321,833	767,772,527	658,400,440

Instead of Lasso: Cross-Validated Forward Selection

Note: ordinal variables now quantitatively encoded and use $\log(\text{response})$

Use 5-fold CV and keep adding variables that lower test RMSE



In Pursuit of a Parsimonious Model

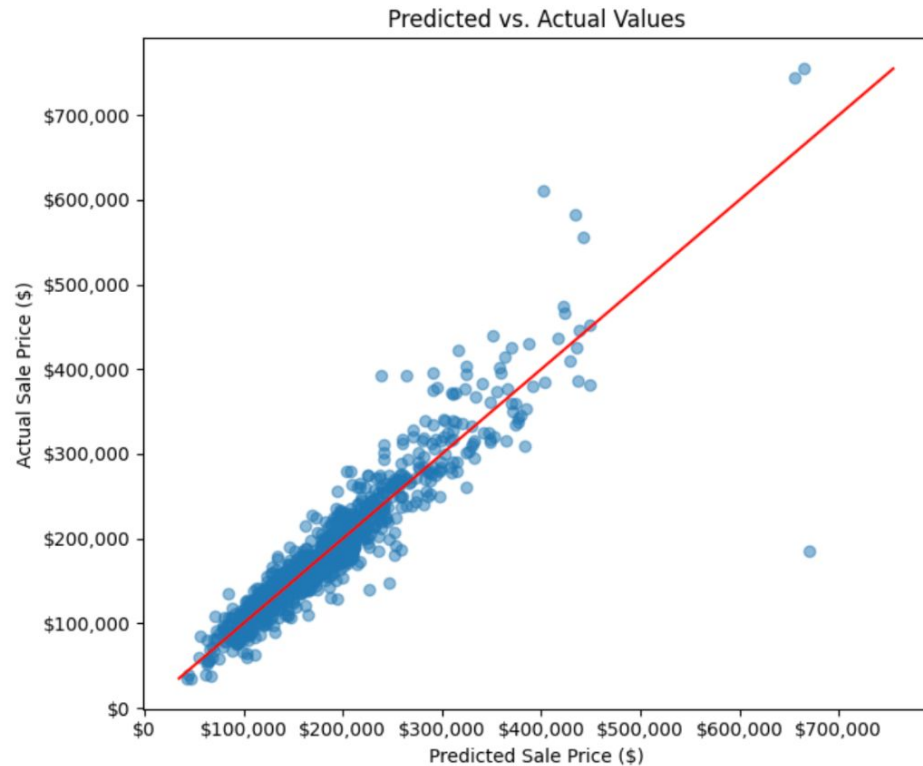
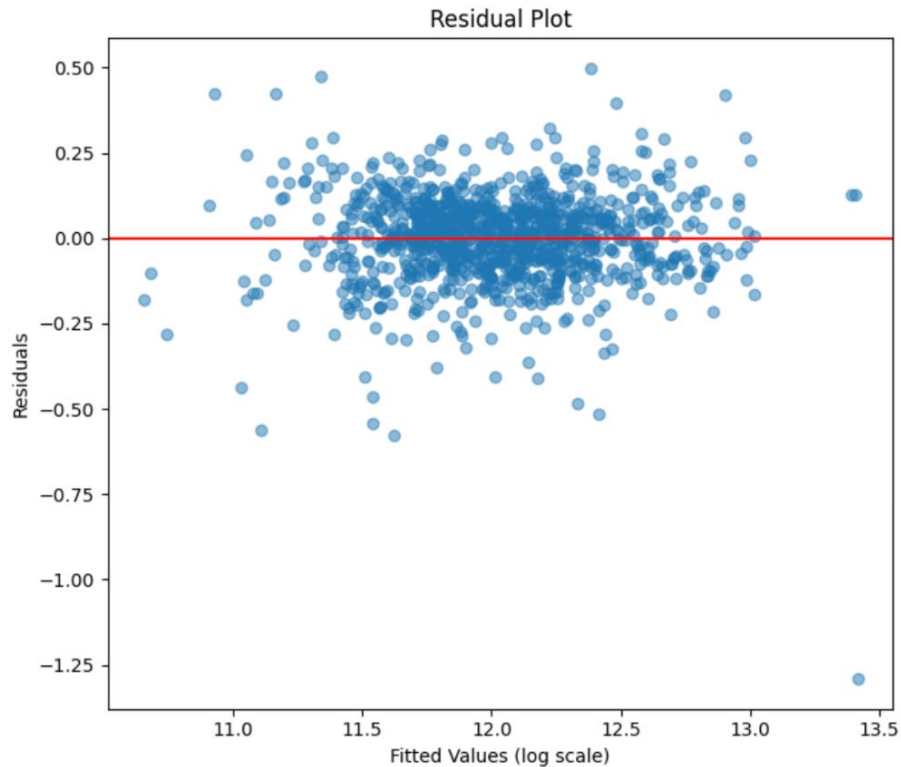
- The forward selection picks 82 out of 205 features, but most of the benefit is accounted for by the first 40
- Starting with the top 40 features, we can iteratively delete those features with p-value significance less than 0.01
 - This gets us down to 32 features
 - But “Functional” and “YearBuilt” have high VIFs (150 and 200)
 - Remove extremely high VIFs, re-run and drop low p-value features, repeat
- Once this process terminates, our model has 28 features
 - We can tolerate insignificant variables if they are levels of otherwise significant variables

Parsimonious Model Final Features

Variables ranked by p-value (most significant first):

	Variable	Coefficient	P-Value
0	GrLivArea	0.000265	2.739787e-104
1	OverallQual	0.075798	4.571900e-42
2	RoofMatl_ClyTile	-1.651571	1.585694e-30
3	GarageCars	0.083389	6.749773e-29
4	MSZoning_C (all)	-0.523973	5.785907e-27
5	MSZoning_RM	-0.128340	2.396959e-22
6	OverallCond	0.029479	8.779530e-13
7	Neighborhood_NridgHt	0.138734	7.848951e-12
8	CentralAir	0.117822	4.736574e-10
9	HeatingQC	0.029397	8.178895e-09
10	MSSubClass_160	-0.124796	1.570918e-08
11	BsmtQual	0.036321	5.341626e-08
12	BsmtExposure	0.022797	4.250819e-07
13	MSZoning_FV	0.101857	4.211314e-06
14	BsmtFullBath	0.044408	8.911581e-06
15	Exterior1st_BrkFace	0.100848	1.035490e-05
16	BsmtFinType1	0.011186	1.716291e-05
17	Neighborhood_Edwards	-0.066918	5.214235e-05
18	WoodDeckSF	0.000143	6.599767e-05
19	Condition1_Artery	-0.084503	2.063358e-04
20	ScreenPorch	0.000261	4.846895e-04
21	Neighborhood_StoneBr	0.138698	7.554391e-04
22	FireplaceQu	0.009323	9.551008e-04
23	LotConfig_CulDSac	0.056607	1.183748e-03
24	SaleType_ConLD	0.157343	1.269193e-03
25	Neighborhood_Crawfor	0.072315	1.602407e-03
26	Heating_GasW	0.098620	7.073336e-03
27	MSZoning_RH	-0.052356	1.509684e-01

Parsimonious Linear Model (28 Variables)



Parsimonious Performance

Test Set Performance Metrics:

Bias	\$-493.26
Maximum Deviation	\$118,296.55
Mean Absolute Deviation	\$16,572.77
MSE	\$542,117,508.27
RMSE	\$23,283.42
MAPE	9.52%
R^2 (Original Scale)	0.9108

Random Forest & Elastic Net

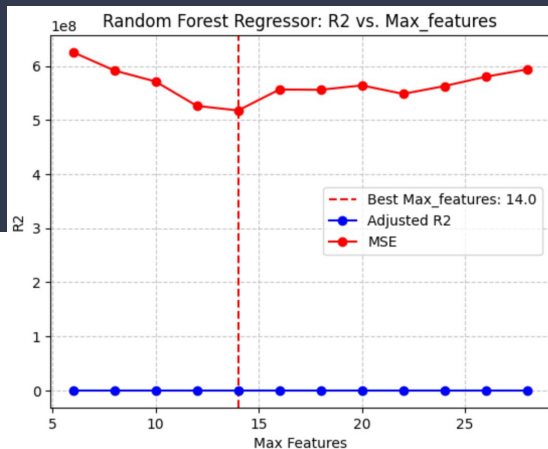
- Random Forest and Elastic Net can perform well for house price prediction because they can handle non-linear relationships and multicollinearity which appear in the dataset.
- Features reduction was done by dropping heavily correlated features ($|r| > 0.75$) and dropping features with $p > 0.20$.
- Missing numerical or nominal values were replaced with the median or mode respectively. All data was scaled after encoding.

Step		Encoding	Multicollinearity	p-value
Feature Size	82	212	119	44

Random Forest

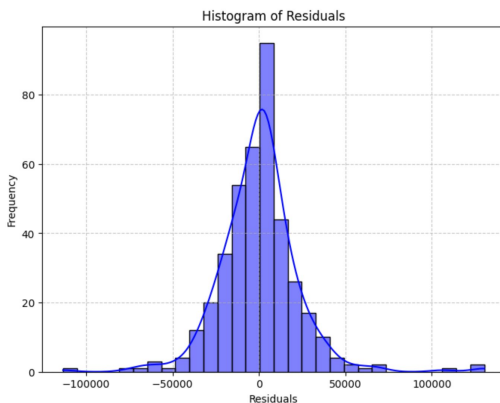
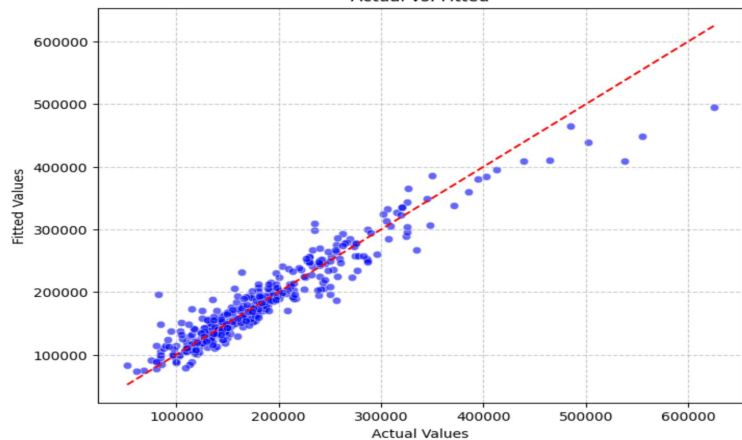
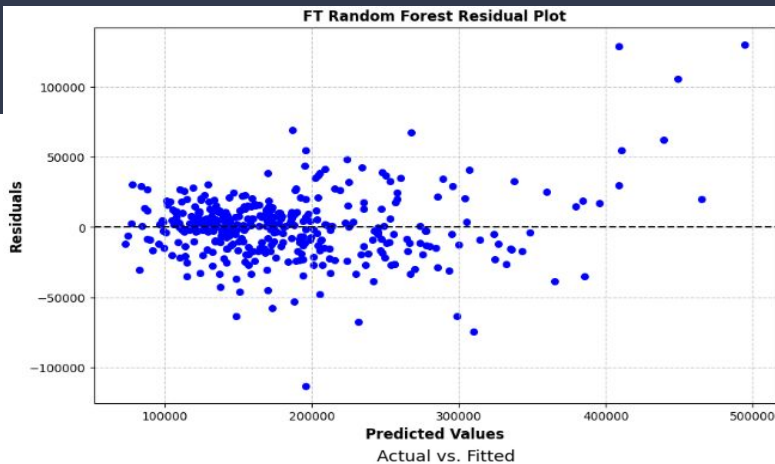
- Random forest is an ensemble learning method that combines multiple decision trees to predict continuous values. It creates multiple subsets of the training data and fits a regression tree to each subset, splitting data at nodes based on features that minimize error.
- Fine-tuning RFR was done by first evaluating its main parameter 'Max_features' and finding which value maximizes its MSE.
- 5-fold cross validation along with a grid search was used to optimize four other parameters that could potentially improve its performance.

* Random Forest(n_estimators=200, min_samples_split=3, min_samples_leaf=2, max_depth=20, max_features=14)



Feature	Importance
OverallQual	0.287
GrLivArea	0.171
GarageCars	0.125
TotalBsmtSF	0.082
YearBuilt	0.078

Random Forest Performance



Metric	Score
R2	0.911
Bias	-101
Max Dev	130,432
MAD	15,655
MSE	538,531,741

Elastic Net

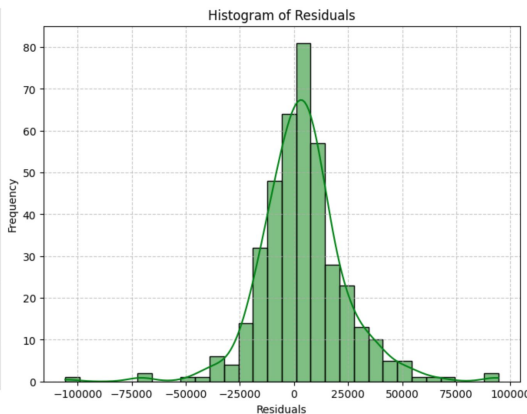
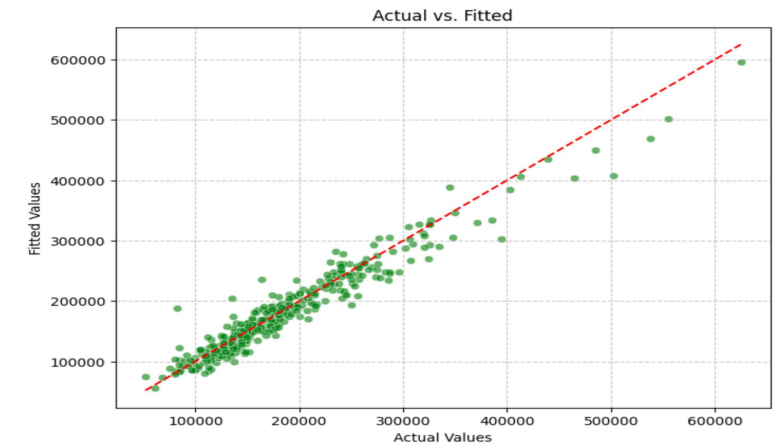
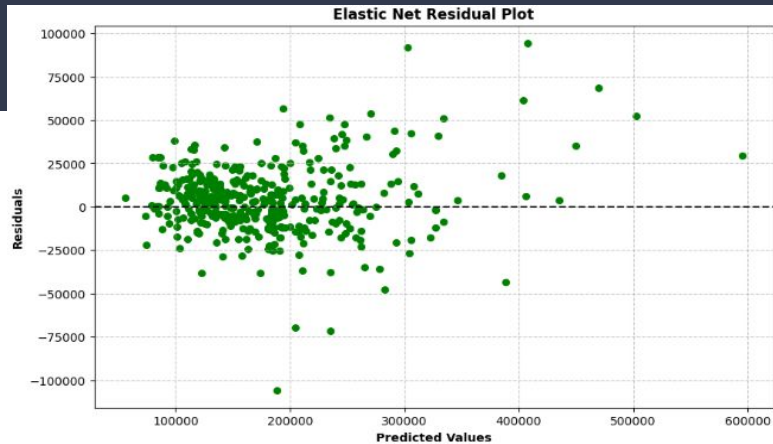
- Elastic Net is a linear regression model that combines the penalties of both Ridge and Lasso regression, making it effective for handling multicollinearity while also performing variable selection.
- The log of the Sale Price was used as the target variable to reduce skewness, which helps improve metrics by making the target distribution more normal and stabilizing variance.

Step		Elastic Net
Feature Size	44	29

*Elastic Net parameters were obtained using 5-fold cross validation ($\alpha=0.01$, l1-ratio=0.5)

Feature	Coefficient
GrLivArea	0.131
OverallQual	0.103
ClyTile	-0.08
YearBuilt	0.06
GarageCars	0.05

Elastic Net (Using Log(SalePrice))



Metric	Score
R2	0.933
Bias	-3889
Max Dev	105,752
MAD	14,223
MSE	409,423,926

Elastic Net > Random Forest ?

- Elastic Net tends to handle multicollinearity between features better due to both L1 and L2 regularization being used which shrink and selects features effectively.
- Random Forest can struggle with high-dimensional sparse data or when overfitting becomes a concern due to many irrelevant or noisy predictors.

The End



Appendix Slides

Shapiro Wilk Normality Test

(kinda explained)

$$H_0 : X \sim N(\mu, \sigma^2)$$

$$H_A : X \not\sim N(\mu, \sigma^2)$$

Hypothesis Testing

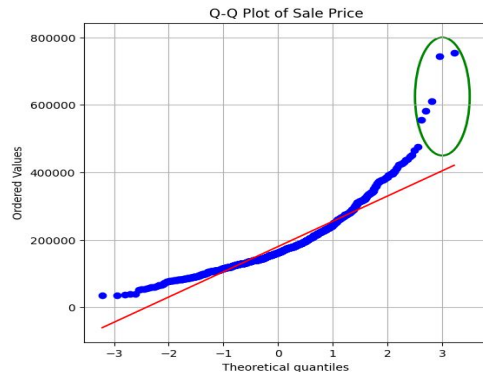
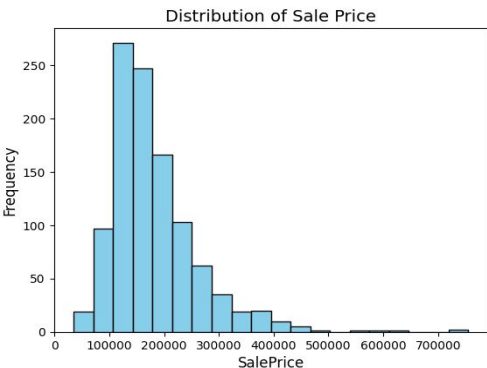
$$W = \frac{(\sum_{i=1}^n a_i x_{(i)})^2}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

Test Statistic Formula

- Visual is OK, but should test
- Numerator is the squared weighted sum of sorted x-values.
- Denominator measures sum of squared deviations
- W around 1 means x is close to normal
- We can get non-significant p-values when working with large datasets and will see that this test was a bad choice in the next few slides

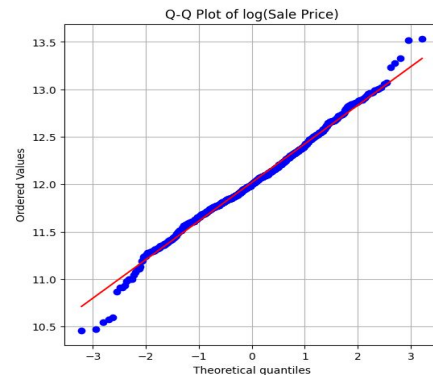
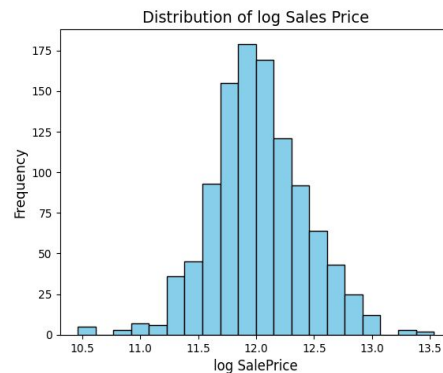
W (test statistic): 0.87343, p-value: 0.0
Sale Price is not Normally distributed.

Normality Comparison of Sale Price vs log(Sale Price)



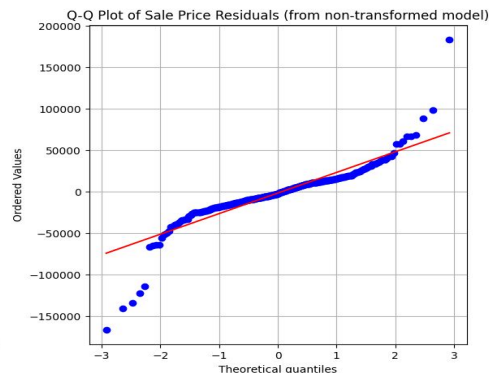
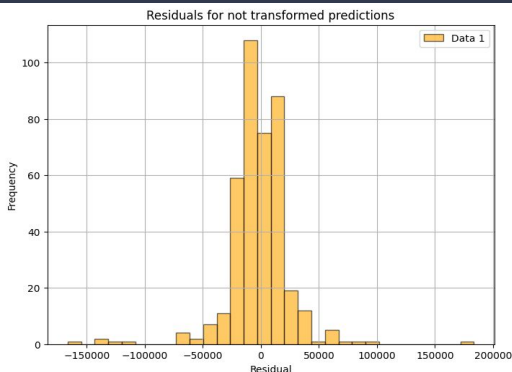
W (test statistic): 0.87343, p-value: 0.0

	Histogram	QQ Plot	SW Test
Sale Price	Not normal	Not normal	Not normal
log(Sale Price)	Normal	Normal	Not normal



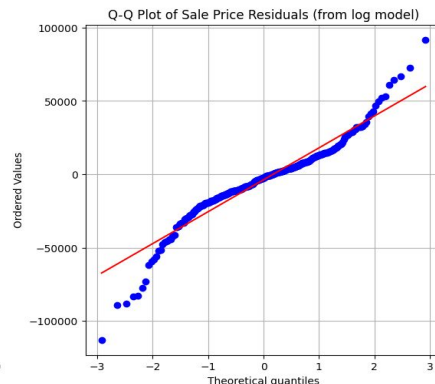
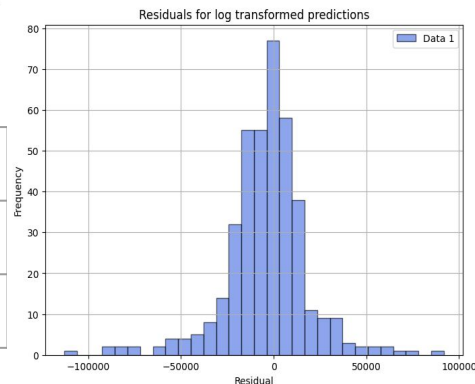
W (test statistic): 0.99143, p-value: 1e-05

Normality of Model Residuals for LASSO



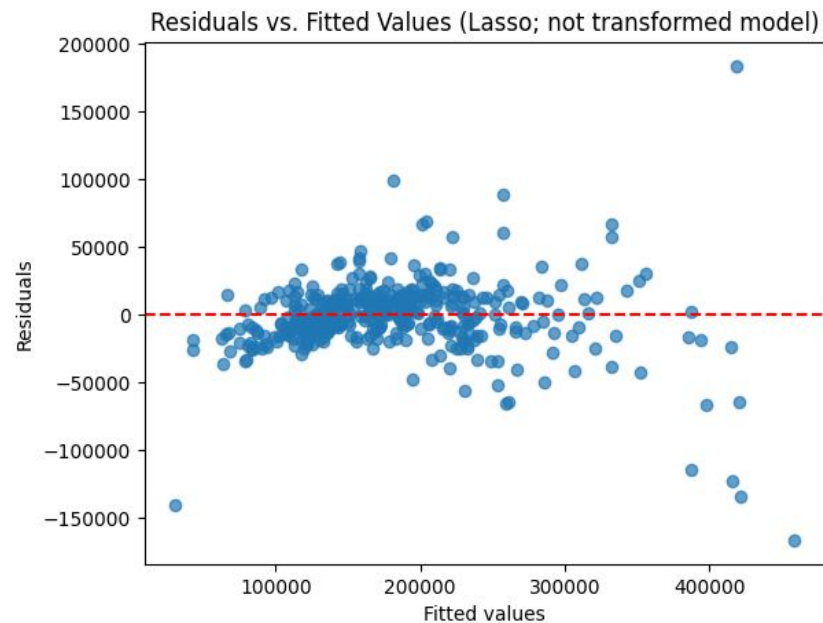
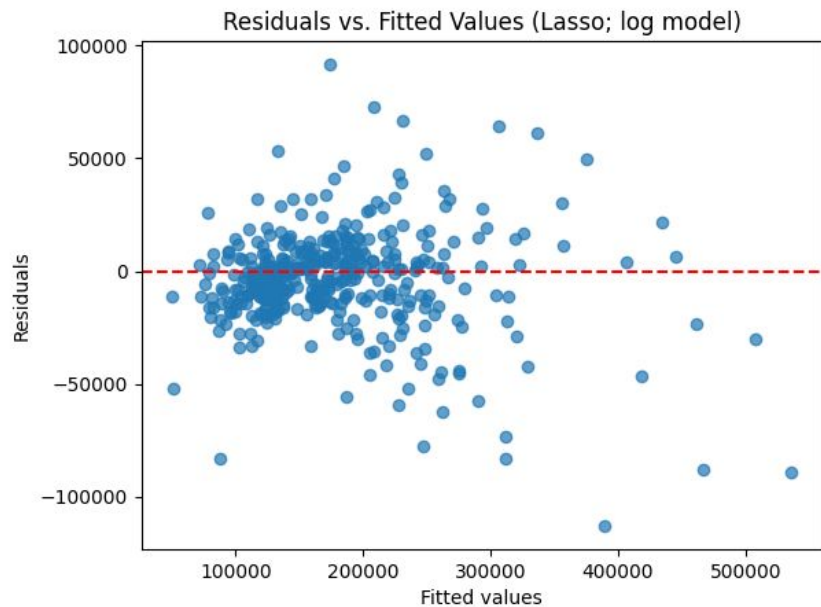
W (test statistic): 0.83064, p-value: 0.0

LASSO Residuals	Histogram	QQ Plot	SW Test
Sale Price	Normal	Maybe	Not Normal
log(Sale Price)	Normal	Maybe	Not Normal



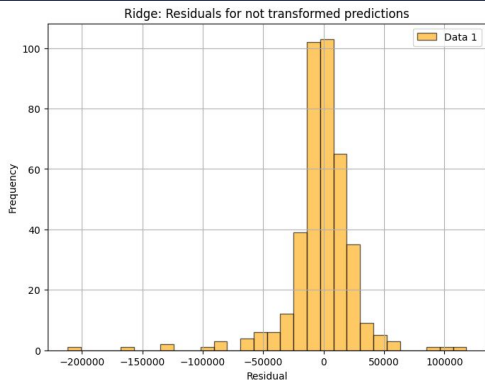
W (test statistic): 0.92698, p-value: 0.0

Model #1: More LASSO residuals



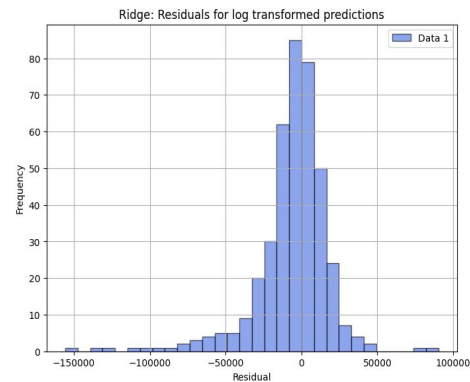
Heteroskedasticity?

Normality of Model Residuals for Ridge



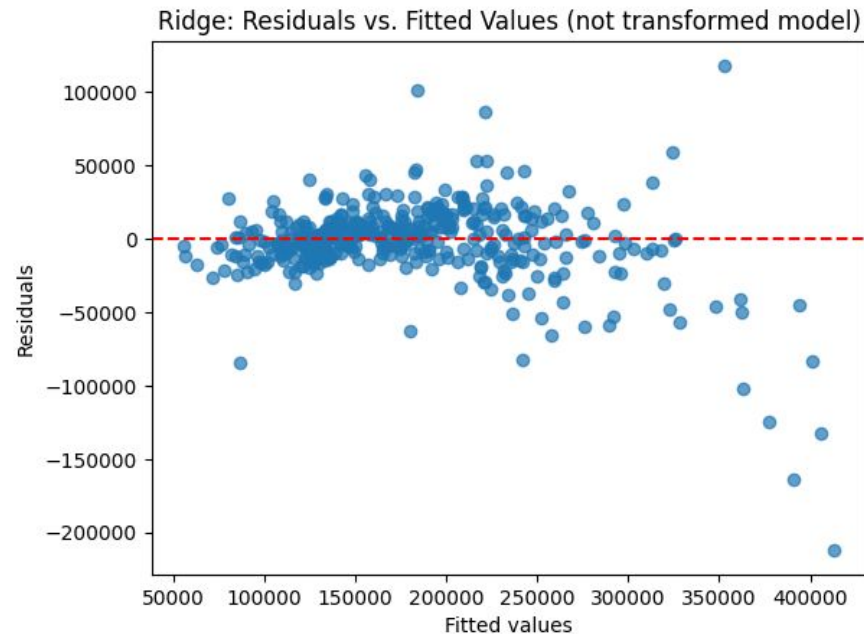
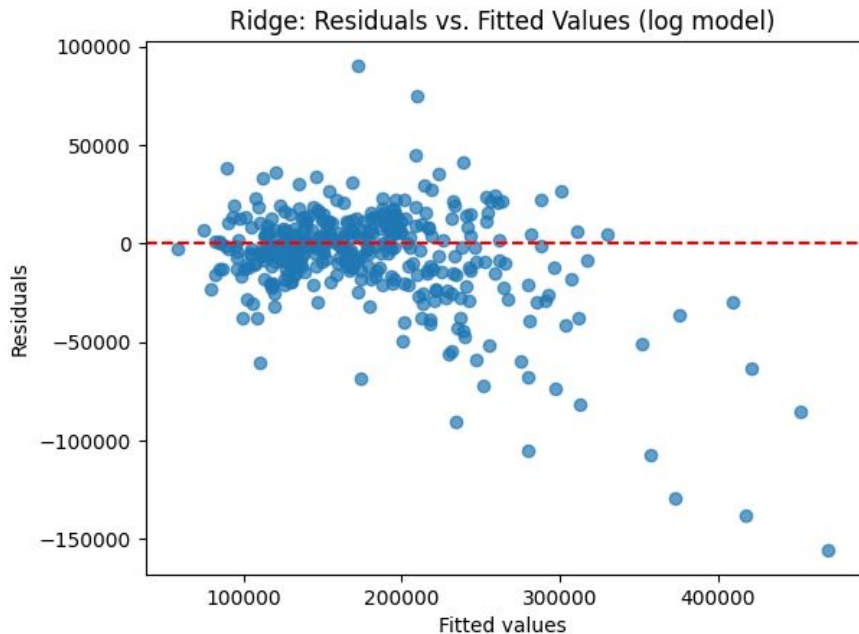
W (test statistic): 0.81223, p-value: 0.0

<i>Ridge Residuals</i>	Histogram	QQ Plot	SW Test
Sale Price	Not Normal	Maybe	Not Normal
log(Sale Price)	Not Normal	Maybe	Not Normal



W (test statistic): 0.85145, p-value: 0.0

Model #1: More Ridge residuals



Heteroskedasticity?

Model #1: LASSO & Ridge (Background I)

There are many possible norms on $\mathbb{R}^n$.

Examples: consider a vector $\mathbf{x}$ in $\mathbb{R}^n$ with elements $\mathbf{x} = (x_1, x_2, \dots, x_n)$. Then:

- 1 The Euclidean distance, or l_2 norm:

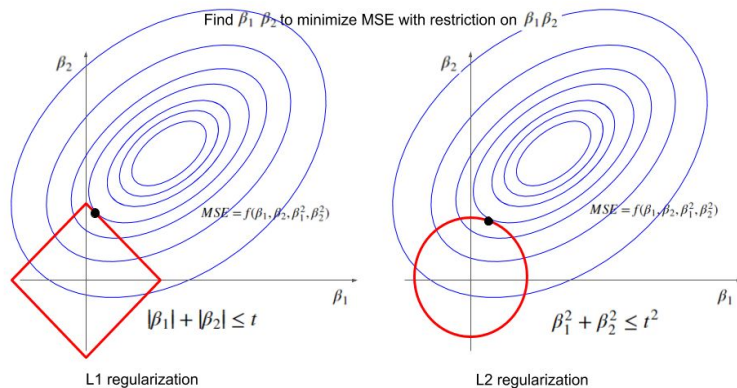
$$\|\mathbf{x}\|_2 = \sqrt{\sum_{i=1}^n x_i^2}$$

- 2 The l_∞ or *sup* norm:

$$\|\mathbf{x}\|_\infty = \max_{1 \leq i \leq n} |x_i|.$$

- 3 The l_1 norm:

$$\|\mathbf{x}\|_1 = \sum_{i=1}^n |x_i|$$



**Least Absolute Shrinkage and Selection Operator (left)
Ridge (right)**

Model #1: LASSO & Ridge (Background II)

Recall, for ridge regression, we seek to find $\tilde{\beta}$ that, for some fixed λ , minimizes

$$\|Y - X\beta\|^2 + \lambda\|\beta\|^2$$

which turns out to be

$$\tilde{\beta} = (X'X + \lambda I)^{-1}X'Y$$

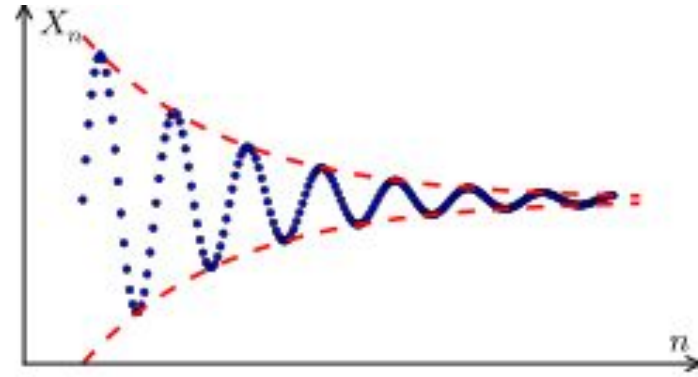
$$\sum_{i=1}^n (y_i - \sum_j x_{ij}\beta_j)^2 + \lambda \sum_{j=1}^p |\beta_j|$$

OLS part

LASSO part

Model #1: LASSO & Ridge (Background III)

- Banach spaces
- Hilbert spaces
- Inner Products
- Orthogonality & Projection
- $\mathbb{R}^n$ with the 2-norm
- Lack of closed form solutions for other norms



Cauchy sequence example